

Probabilistic Forecasting of Regional Redispatch Volumes in the German Electricity Market

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Redispatch measures play a key role in managing grid congestion in European electricity markets. In Germany, redispatch volumes have increased in the recent years due to the rapid expansion of renewables and transmission bottlenecks. Estimating these volumes is increasingly important with the planned Redispatch 3.0 framework, which introduces market-based components. We develop probabilistic day-ahead forecasts of regional redispatch volumes using publicly available data. Four redispatch regions with consistent redispatch behavior are identified via clustering. For each region, we compare an autoregressive model with bootstrapped prediction intervals to a transformer-based neural network with quantile regression. Results show that the autoregressive model performs better in highly volatile regions, while the transformer yields more accurate forecasts in stable regions. Our data-driven forecasting framework offers insights for congestion management and bidding in electricity markets.

Introduction

Efficient electricity market design is essential for ensuring secure grid operation and for providing the right incentives for market participants. In the United States, transmission system operators (TSOs) have transitioned to nodal pricing (locational marginal pricing) starting in 1999, where electricity prices vary by location and reflect both generation costs and transmission constraints. The earlier zonal pricing system, which used uniform prices across large regions, often led to market-clearing schedules that were infeasible given the grid constraints, requiring costly redispatch interventions to adjust generation schedules and relieve network congestions. The experience in markets such as ERCOT and PJM showed that nodal pricing can reduce redispatch volumes, align economic signals with network constraints, and improve market efficiency (Triolo and Wolak, 2022; Harvey and Hogan, 2010).

In contrast, the coupled European electricity market operates under a zonal pricing regime, where day-ahead prices are uniform within national or regional bidding zones. While zonal pricing simplifies trading and market clearing, it does not explicitly account for intra-zonal network constraints. Consequently, when market-clearing schedules result in physically infeasible flows, TSOs intervene through redispatch measures to relieve network bottlenecks. This requires instructing specific power plants to change their operating schedules either by increasing (upward or positive redispatch) or decreasing (downward or negative redispatch) their planned output. The amount of generation capacity adjusted to address congestion is referred to as redispatch volumes.

In Germany, redispatch volumes have increased over the years due to the country's rapid expansion of renewable energy, particularly wind power in the north, and the high electricity demand in the south (Titz et al., 2024). This geographic imbalance and the insufficient transmission network capacity cause frequent congestions and have led Germany to record the highest redispatch volumes and costs in Europe (Thomassen et al., 2024). Germany currently operates under a cost-based redispatch regime known as Redispatch 2.0. Under this system, TSOs hold contractual rights to modify the output of individual generators, who are financially compensated based on declared costs. A planned reform, Redispatch 3.0, seeks to enhance this mechanism by incorporating smaller, decentralized flexibility units and introducing market-based components. Local Flexibility Markets (LFMs) are a key example that enable decentralized procurement of flexibility services and have been successfully integrated in countries such as the United Kingdom (Piclo Flex) and the Netherlands (GOPACS) (Michaelis et al., 2025). In LFMs, flexibility providers, including distributed energy resources and demand-side participants, can offer capacity to increase or reduce consumption or generation within the geographic area defined by the market.

The integration of market-based components into Redispatch 3.0 creates new opportunities around redispatch, increasing the need for information systems that provide market participants with data-driven insights to develop bidding strategies. Accurate forecasts of redispatch volumes allow participants to anticipate locational congestion and adjust their strategies accordingly. For instance, if a generation unit expects higher upward redispatch in a specific region, they might

reduce their day-ahead market bids and instead position themselves to participate profitably in the redispatch market. Such strategic behavior, known as inc-dec gaming, has been identified as an important theoretical concern in the design of LFM (Ehrhart et al., 2022), and relies on accurately predicting the regional redispatch volumes. Forecasting at the regional rather than individual-unit level is particularly useful because redispatch decisions are influenced by network topology and regional power flows rather than by the operational characteristics of single units. Previous work on redispatch forecasting (Haarp et al., 2018) has focused on predicting day-ahead whether specific power plants – those with historically high redispatch activity – will be redispatched, without forecasting the actual redispatch volumes. To fill this gap, we develop probabilistic models to forecast regional redispatch volumes that can support market participants in developing bidding strategies under Redispatch 3.0.

Related Work

Energy forecasting has been studied in both the information systems and energy literature. For example, Garcia et al., 2024 propose a framework that jointly optimizes forecasting and decision-making processes for energy applications. Probabilistic forecasting techniques have been used for modeling uncertainty in various energy applications, such as renewable generation (Miraki et al., 2025), load (Hong et al., 2016), and day-ahead electricity prices (Munian and Ziel, 2020). While these applications focus on market inputs, much less attention has been given to forecasting corrective actions taken after market clearing - such as redispatch. One exception is the work by Haarp et al. (2018), who use a feed-forward neural network and an extra-tree classifier to predict, on a binary basis, whether individual power plants will be redispatched the next day. However, to the best of our knowledge, no prior work has addressed the problem of forecasting redispatch volumes, particularly at a regional level.

Contributions

We identify spatiotemporal patterns in German redispatch by analyzing public unit-level data from 2021-2024. Using a clustering approach, we delineate four distinct regions that vary in redispatch direction, average levels, and variability. In the second step, we develop probabilistic forecasting

models to predict day-ahead redispatch volumes in each identified region. To estimate the conditional distributions of redispatch volumes, we implement and compare an autoregressive model with bootstrapped prediction intervals and a transformer-based neural network architecture with quantile regression. Our results show that in regions with high redispatch variability, the autoregressive model achieves better predictive performance, while the transformer architecture is more effective in regions with lower variability. Overall, our study contributes a data-driven framework for regional redispatch forecasting that can inform decision-making for electricity market participants, support bidding under Redispatch 3.0, and enhance operational planning for TSOs.

Spatiotemporal Redispatch Patterns

In this section, we describe our approach for identifying spatiotemporal redispatch patterns. We use the time series of redispatch volumes reported on [Netztransparenz.de](https://www.netztransparenz.de), covering the period from January 1, 2021, to December 31, 2024. Our analysis focuses on power-related redispatch actions, excluding countertrading measures and redispatch test procedures. Since the dataset does not include geographic coordinates for the redispatched units, we inferred their locations using the list of power plants provided by the Open Power System Data platform, which contains coordinate information. We matched unit names in both data sets using fuzzy string matching based on the Levenshtein distance. To compare two strings, we computed a weighted similarity score based on the string lengths and considered a threshold of 90 to classify the strings as similar (the score is between 0 and 100). Units that could not be matched using this approach were geolocated manually. In total, we consider 264 redispatched units.

To identify regions with consistent redispatch volumes patterns, we clustered the units using K-Means with their average redispatch volume in 2021-2024 and geographic coordinates (latitude and longitude) as features. Our analysis reveals that the yearly redispatch volumes and the number of redispatched units have increased between 2021 and 2024, but the spatial patterns remained consistent. Specifically, northern Germany is consistently characterized by downward redispatch, whereas the southern regions exhibit persistent upward redispatch. This behavior is observed in Figure 1, where a positive (negative) value indicates upward (downward) redispatch. This spa-

tial redispatch asymmetry reflects the structural imbalance in Germany's power system: the north hosts a high concentration of renewable generation – particularly onshore and offshore wind – while the south has higher electricity demand, including industrial consumption and reduced local generation capacity following nuclear and coal plant retirements. As a result, power frequently needs to be curtailed in the north and ramped up in the south to maintain grid stability. We argue that clustering units based on their mean redispatch volume (and not the whole time series) and location effectively captures these regional patterns without substantial information loss. The optimal number of clusters was determined using the Elbow method based on the inertia metric, which suggested an optimal clustering into four regions (Figure 1). The evolution of total daily redispatch volumes in each region is depicted in Figure 2.

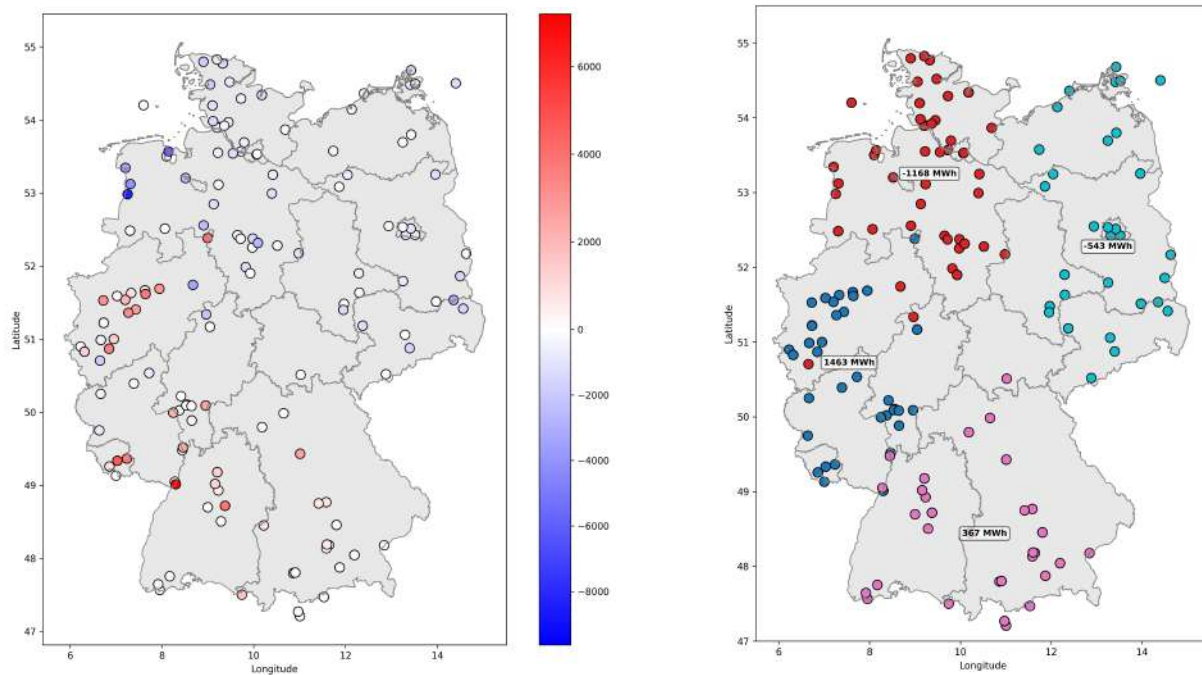


Figure 1: Mean redispatch volumes (MWh) per unit in 2021-2024 (left) and regions identified with K-Means (right). For each region, the mean redispatch volume is shown.

The identified regions exhibit distinct and consistent redispatch patterns. The red and turquoise regions are characterized by downward redispatch, with the red region showing a substantially higher average volume than the turquoise region (-1,168 MWh vs. -543 MWh). In contrast, the

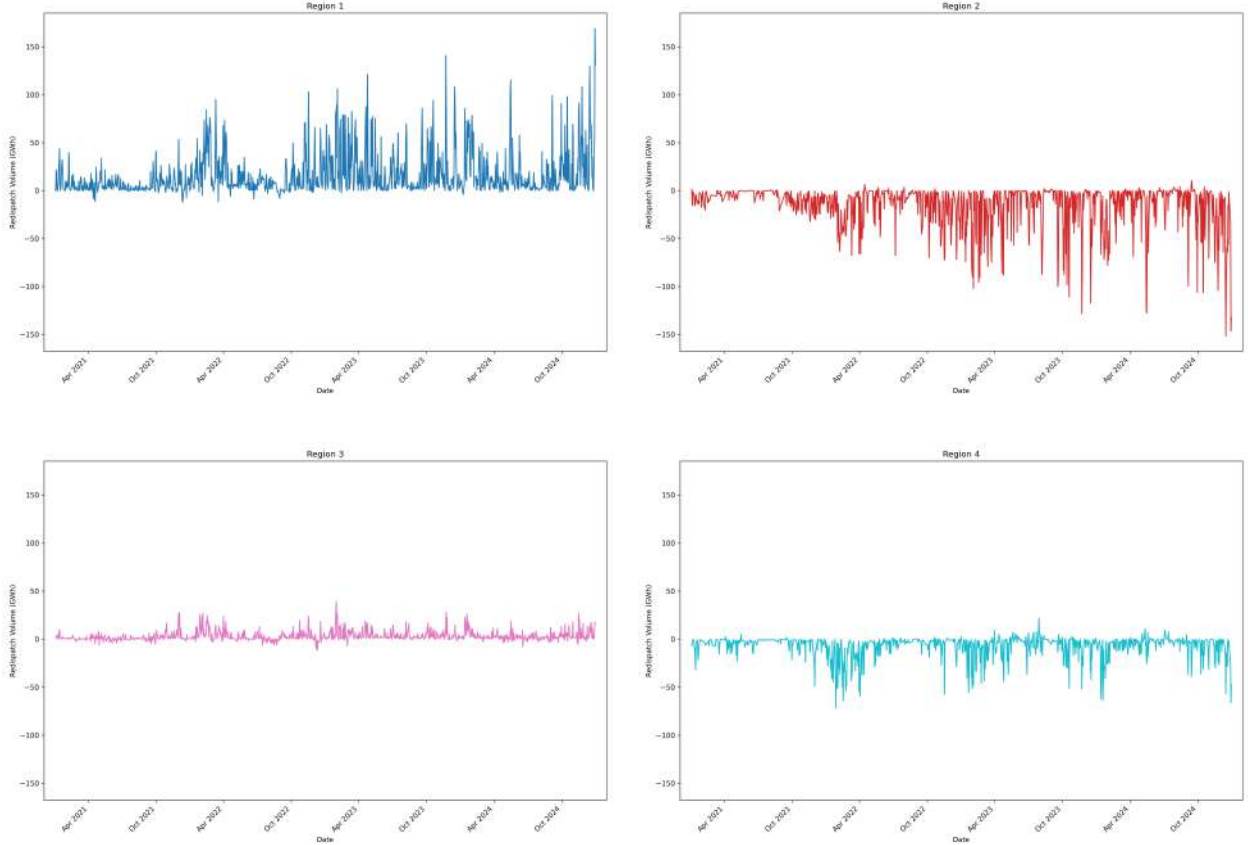


Figure 2: Evolution of total daily redispatch volumes in each region between 2021 and 2024.

blue and pink regions display upward redispatch, which is more pronounced in the blue region (1,463 MWh vs. 367 MWh). As expected, the upward and downward redispatch volumes tend to mirror each other, indicating that congestion is typically solved by increasing dispatch at one set of plants while reducing dispatch at another. Moreover, the redispatch volumes tend to be higher during the winter months compared to summer. In the following sections, the regions are numbered from 1 to 4 in the order: blue, red, pink, turquoise.

Forecasting Models

In what follows, we present the models we implemented to forecast day-ahead redispatch volumes in each identified region. The code will be published upon publication of the paper in a Github repository. We first state the forecasting problem that we consider.

Given a set of days T , let $\{y_t\}_{t=1}^T$ denote the time series of redispatch volumes and let $\mathbf{x}_t \in \mathbb{R}^d$

denote the vector of explanatory variables, including the lagged values of y and exogenous variables. Based on information available up to day t , our goal is to generate a day-ahead forecast $\hat{y}_{t+1}$ for the redispatch volume on the next day $t + 1$.¹ One option for doing this is to estimate the conditional mean of y_{t+1} given $\mathbf{x}_t$, i.e., $\hat{y}_{t+1} = \mathbb{E}[y_{t+1} \mid \mathbf{x}_t]$. In this paper, besides identifying the point estimate $\hat{y}_{t+1}$, we employ probabilistic forecasting to estimate the entire conditional distribution $F_y(y_{t+1} \mid \mathbf{x}_t)$ of redispatch volumes given $\mathbf{x}_t$. This approach allows us not only to predict the most likely future redispatch volume but also to quantify the uncertainty associated with this forecast. Based on the estimated conditional distribution, we construct a prediction interval as follows:

$$\text{PI}_{1-\alpha}(\mathbf{x}_t) = \left[Q_y\left(\frac{\alpha}{2} \mid \mathbf{x}_t\right), Q_y\left(1 - \frac{\alpha}{2} \mid \mathbf{x}_t\right) \right]$$

where $Q_y(\alpha \mid \mathbf{x}_t)$ represents the conditional quantile function of y_{t+1} given $\mathbf{x}_t$ at level $\alpha \in (0, 1)$, i.e., $Q_y(\alpha \mid \mathbf{x}_t) = \inf\{y : F_y(y \mid \mathbf{x}_t) \geq \alpha\}$. This prediction interval provides a range within which we expect the future redispatch volume to fall with probability $1 - \alpha$, given $\mathbf{x}_t$.

We consider the generation, load, and day-ahead prices for Germany as exogenous variables (obtained from the ENTSO-E Transparency Platform). Our analysis shows that the redispatch volumes in regions 2 and 4 (1 and 3) have similar correlation patterns, being negatively (positively) correlated with generation and load, and positively (negatively) correlated with day-ahead prices.

Autoregressive Model with Bootstrapped Intervals

The first model we employ is an Autoregressive Integrated Moving Average with Exogenous Variables (ARIMAX) model. An ARIMAX model of order (p, d, q) is defined as:

$$y_t = c + \beta^\top \mathbf{x}_t + \sum_{j=1}^q \psi_j \varepsilon_{t-j} + \varepsilon_t, \quad (1)$$

where $\mathbf{x}_t = [y_{t-1}, \dots, y_{t-p}, x_{t,1}, \dots, x_{t,d-p}]^\top$, y_t denotes the original series, c is a constant, β, ψ are model parameters, and ε_t is the residual term. We performed the Augmented Dickey-Fuller test which suggested that we can treat the time series of redispatch volumes in each region as a

¹Extending the forecasting horizon to weekly or monthly scales could be valuable for strategic planning and is left for future work.

stationary process, i.e., we have $d = 0$. In Equation (1), the term $\beta^\top \mathbf{x}_t$ represents the autoregressive and exogenous components combined into a single linear predictor, while the sum corresponds to the moving average component. Intuitively, the model performs a regression in which the lagged values $[y_{t-1}, \dots, y_{t-p}]$ and the exogenous variables are the inputs and $\{y_t\}_{t=2}^T$ are the outputs. We employ an ARIMAX model to forecast redispatch volumes due to its simplicity and interpretability with respect to the explanatory variables.

We identify the optimal order of the ARIMAX models by selecting the parameter combination that yields the lowest root mean squared error (RMSE) on the training set, using a grid search procedure. To reduce the risk of overfitting, we restrict the search space for the parameters p and q to values between 0 and 5. The best parameters are then used to generate prediction intervals using a bootstrap approach that consists of the following steps: (1) Estimate the ARIMAX model on historical data and obtain residuals ε_t . (2) Generate an initial point forecast for the test period using the fitted model. (3) Draw bootstrap samples of residuals and add them to the point forecast to simulate future redispatch scenarios. (4) Repeat the bootstrap process 1000 times to generate an empirical distribution of forecasts. (5) Compute the desired lower and upper quantiles of the simulated forecasts to construct the prediction interval. We report the median of the bootstrapped forecast distribution as our point forecast, which ensures consistency with the bootstrapped prediction intervals and provides a robust central estimate that is less affected by skewness or extreme values in the forecast distribution. Our implementation is based on the statsmodels library.

Transformer with Quantile Regression

The second model we employ is based on a transformer architecture. Transformers are neural network (NN) architectures that model relationships in sequential data using self-attention mechanisms (Vaswani et al., 2017). Unlike recurrent networks, which process sequences step by step and may struggle with long-range dependencies, transformers leverage self-attention to capture temporal relationships simultaneously and efficiently. Originally developed for natural language processing, they have also been successfully applied to time series forecasting (Wen et al., 2023).

Given a lookback window of length L consisting of input feature vectors $\{\mathbf{x}_t\}_{t=1}^L$, with $\mathbf{x}_t \in \mathbb{R}^d$,

the transformer first projects each $\mathbf{x}_t$ into a higher-dimensional space, i.e., $\tilde{\mathbf{x}}_t = W_{\text{in}} \mathbf{x}_t + \mathbf{b}$, where $W_{\text{in}} \in \mathbb{R}^{d_{\text{model}} \times d}$. Since transformers do not have inherent sequence order, a positional encoding PE_t is added: $\mathbf{z}_t^{(0)} = \tilde{\mathbf{x}}_t + \text{PE}_t$, resulting in the initial embedding matrix $Z^{(0)}$.² This matrix is then processed by a stack of transformer encoder layers. Each of these performs four operations. First, in the multi-head self-attention operation, each time step in the window computes attention scores with all other time steps, allowing the model to learn dependencies across the entire window:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V,$$

where $Q = Z^{(\ell-1)} W_Q$, $K = Z^{(\ell-1)} W_K$, $V = Z^{(\ell-1)} W_V$ are the query, key and value matrices, respectively, and $W_Q, W_K, W_V \in \mathbb{R}^{d_{\text{model}} \times d_k}$ are learned weight matrices. Intuitively, the dot products $QK^\top$ measure similarity between time steps, and the softmax transforms these scores into weights indicating how much each time step should contribute to the representation of the others. Multi-head attention repeats this process in parallel, allowing the model to capture different types of relationships. Second, the output of the attention mechanism is combined with the original input through a residual connection and then normalized using layer normalization:

$$H^{(\ell)} = \text{LayerNorm} \left(Z^{(\ell-1)} + \text{Attention output} \right)$$

Third, the normalized output passes through a feedforward NN (FFN) applied to each time step, and, finally, another residual connection and layer normalization are applied:

$$\text{FFN}(H^{(\ell)}) = \text{ReLU}(H^{(\ell)} W_1 + b_1) W_2 + b_2, \quad Z^{(\ell)} = \text{LayerNorm} \left(H^{(\ell)} + \text{FFN}(H^{(\ell)}) \right)$$

After the final encoder layer, the representation of the last time step in the window $\mathbf{h}_t = Z_t^{(L)}$ is selected and passed to a deep feedforward NN to perform quantile regression, which estimates $Q_y(q \mid \mathbf{x}_t)$. Our implemented transformer architecture is detailed in Appendix A. The model is trained using the pinball loss function³ and forecasts three quantiles q_1, q_2, q_3 simultaneously, out-

²The positional encoding PE_t provides information about the relative position of each time step, typically using sinusoidal functions (see Appendix A).

³Given a quantile q , ground truth y and forecast $\hat{y}$, the pinball loss function is $\mathcal{L}_q(y, \hat{y}) = (y - \hat{y}) (q - \mathbf{1}_{\{y < \hat{y}\}})$.

putting a three-dimensional vector $\hat{\mathbf{y}}_t = (\hat{y}_t^{(q_1)}, \hat{y}_t^{(q_2)}, \hat{y}_t^{(q_3)})$. We let $q_2 = 0.5$ and use the median forecast as our point estimate $\hat{y}_t^{(0.5)}$. A prediction interval at nominal coverage level $1 - \alpha = q_3 - q_1$ is given by $\text{PI}_{1-\alpha}(\mathbf{x}_t) = [\hat{y}_t^{(q_1)}, \hat{y}_t^{(q_3)}]$. Thus, the interval $\text{PI}_{1-\alpha}(\mathbf{x}_t)$ captures the range within which the future redispatch volume y_{t+h} is expected to lie with probability $1 - \alpha$.

Results

Table 1 reports the root mean squared error (RMSE), mean absolute error (MAE), and the empirical coverage for the ARIMAX and transformer models across all regions and considered prediction interval levels. The empirical coverage is defined as the fraction of times the true redispatch value falls within the predicted interval, out of the total number of forecasts, which may differ from the nominal coverage level used when constructing the intervals (column “Interval”). RMSE and MAE are reported for the point estimates. Also, Figure 3 illustrates the average interval widths.

Table 1: Evaluation of results.

Reg.	Interval	RMSE		MAE		Empirical coverage	
		ARIMAX	Transformer	ARIMAX	Transformer	ARIMAX	Transformer
1	70%	22,689.81	24,496.02	12,769.64	14,039.66	0.66	0.64
	80%	22,685.59	25,097.50	12,775.22	14,626.77	0.77	0.55
	90%	22,683.34	24,195.92	12,764.60	14,139.61	0.85	0.83
	98%	22,716.22	23,784.49	12,767.49	13,406.42	0.94	0.95
2	70%	25,075.53	28,030.65	14,208.61	14,895.32	0.71	0.63
	80%	25,058.78	28,341.13	14,212.78	14,647.04	0.74	0.68
	90%	25,068.91	28,518.13	14,248.30	14,518.76	0.84	0.81
	98%	25,066.45	26,605.24	14,229.57	14,262.42	0.94	0.91
3	70%	4,615.30	4,532.55	3,050.32	2,977.13	0.60	0.53
	80%	4,615.94	4,581.42	3,042.20	2,939.82	0.77	0.65
	90%	4,625.41	4,553.16	3,062.29	2,999.08	0.89	0.77
	98%	4,606.14	4,461.61	3,042.33	2,970.75	0.99	0.91
4	70%	9,460.96	9,406.11	5,752.98	4,850.50	0.79	0.73
	80%	9,468.94	9,470.06	5,752.37	4,807.00	0.82	0.77
	90%	9,476.37	9,580.96	5,761.49	4,804.27	0.90	0.87
	98%	9,463.36	9,107.09	5,769.77	4,699.91	0.98	0.91

Overall, the results show a strong dependence of model performance on the redispatch variability of each region. For regions with high redispatch variability, namely regions 1 and 2, the

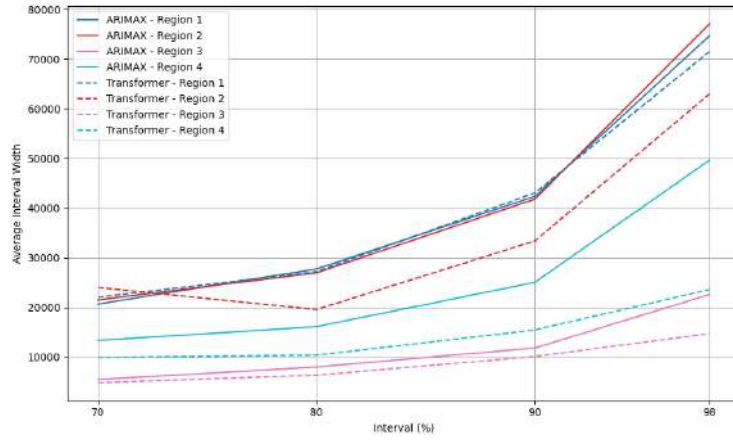


Figure 3: Average interval width vs. interval (%) by model and region.

ARIMAX model outperforms the transformer in point forecast accuracy. For example, in region 1, the ARIMAX achieves lower RMSE values across all interval levels, with an RMSE of 22,716.22 MWh at the 98% interval compared to 23,784.49 MWh for the transformer. This superior performance can be attributed to ARIMAX’s ability to capture sharp peaks and local fluctuations in redispatch volumes through its autoregressive structure and direct incorporation of exogenous variables. Transformers often smooth out sudden spikes because their attention spreads across the whole lookback window, which can weaken signals from short-term redispatch variability.

In contrast, in regions characterized by lower redispatch variability, namely regions 3 and 4, the transformer model leads to better point forecasts. For instance, in region 4, the transformer achieves a lower RMSE of 9,107.09 MWh and a lower MAE of 4,699.91 MWh at the 98% interval, compared to 9,463.36 MWh and 5,769.77 MWh, respectively, for ARIMAX. These improvements can be explained by the transformer’s ability to model longer-term dependencies without overfitting to noise. Thus, in more stable regions where redispatch volumes are less dominated by abrupt spikes, the transformer architecture performs better.

Regarding uncertainty quantification, the ARIMAX model achieves a higher empirical coverage, but produces wider prediction intervals, especially in less volatile regions. This is due to its estimated residual variability and the implemented bootstrapped approach, which propagates uncertainty from historical forecast errors into future predictions. While such wide intervals of-

fer conservative coverage, they may be impractically large for operational decision-making. In contrast, the transformer computes narrower prediction intervals that closely follow the path of its point forecasts and are tight enough to be practically useful, especially in less volatile regions.

Conclusion

We present a probabilistic forecasting framework for day-ahead redispatch volumes in the German electricity market, motivated by the increasing importance of congestion management under Redispatch 3.0. By identifying four distinct redispatch regions using spatial and temporal features, we show that redispatch behavior exhibits strong regional patterns. Our empirical analysis compares an ARIMAX model with bootstrapped intervals and a transformer-based neural network with quantile regression. Results indicate that model performance depends on regional volatility: ARIMAX performs better in volatile regions, while transformers outperform in more stable areas. This study advances the information systems perspective on data-driven decision support by illustrating how predictive information systems can transform public data into actionable insights for congestion management. The proposed forecasting framework offers a foundation for regional bidding strategies and transparency in emerging flexibility markets.

Acknowledgments

The authors gratefully acknowledge financial support from the German Research Foundation (Deutsche Forschungsgemeinschaft, DFG) [BI 1057/10]. Special thanks are extended to Markus Ewert for his valuable comments on this work.

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A Implemented Transformer Architecture

The architecture of our Transformer model is shown in Table 2. In this table, “None” serves as a placeholder for the batch size, L represents the length of the lookback window and d_{in} is the number of input features per time step (4). As positional encoding, sinusoidal encoding was implemented. Eight attention heads were used. The model was trained for 100 epochs with early stopping using the Adam optimizer, with a batch size of 8 and lookback window of size 15. The dataset was split into training, validation, and test sets in a 60-20-20 ratio.

Table 2: Implemented Transformer Architecture

Layer (type)	Output Shape
Input Layer	(None, L , d_{in})
Linear Projection	(None, L , 128)
Positional Encoding	(None, L , 128)
Transformer Encoder Layer $\times 4$	(None, L , 128)
Layer Normalization	(None, L , 128)
Select Last Time Step	(None, 128)
Fully Connected Layer ($128 \rightarrow 256$)	(None, 256)
ReLU Activation	(None, 256)
Dropout (0.2)	(None, 256)
Fully Connected Layer ($256 \rightarrow 128$)	(None, 128)
ReLU Activation	(None, 128)
Dropout (0.2)	(None, 128)
Fully Connected Layer ($128 \rightarrow 64$)	(None, 64)
ReLU Activation	(None, 64)
Dropout (0.2)	(None, 64)
Fully Connected Layer ($64 \rightarrow 3$)	(None, 3)

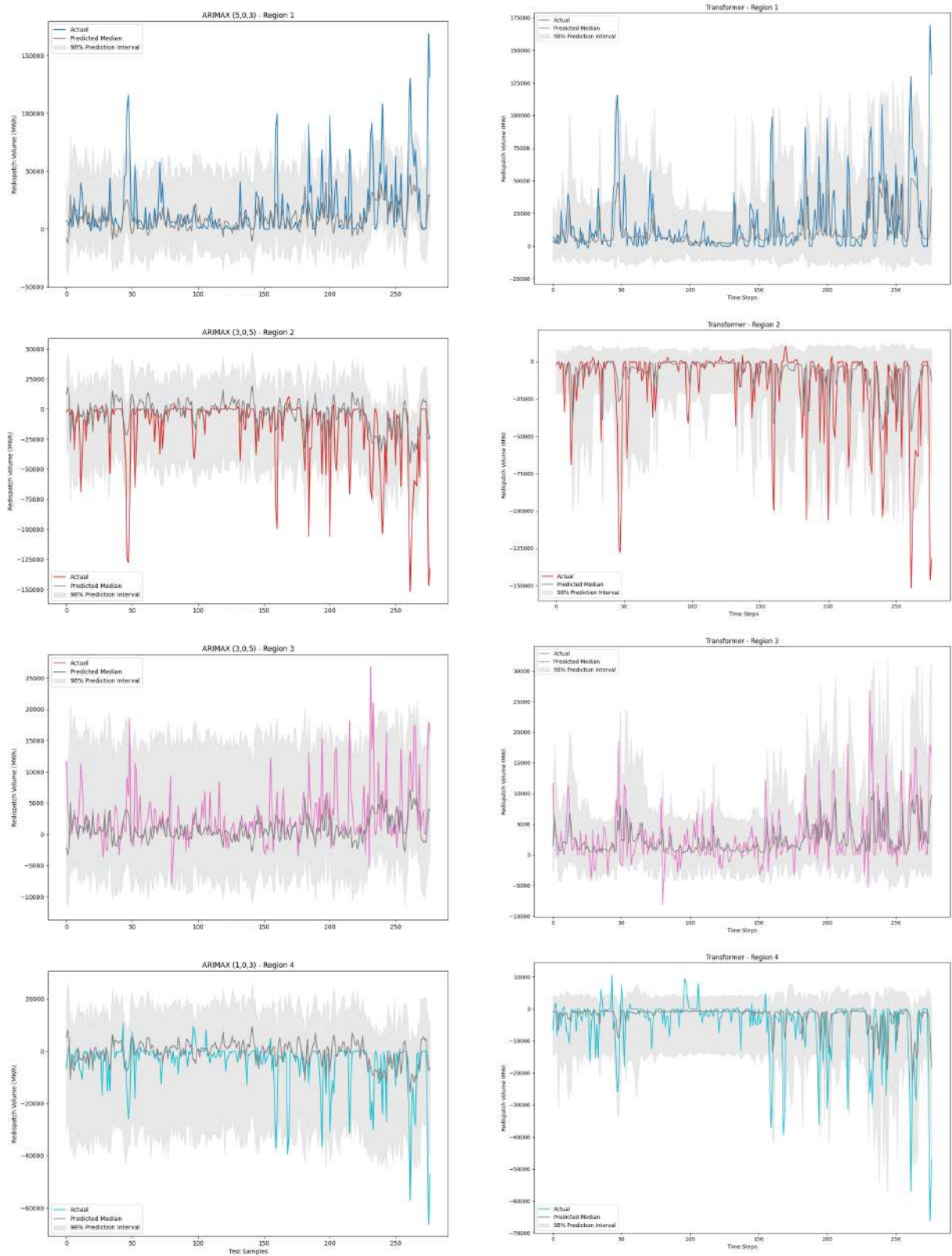


Figure 4: Forecasts per region computed with ARIMAX (left) and transformer models (right).